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Large Language Models, Small Labor Market Effects

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Abstract

We examine the early labor market impacts of AI chatbots by linking large-scale, representative adoption surveys to administrative labor records in Denmark. Using difference-in-differences, we estimate precise null effects on earnings and recorded hours at both the worker and workplace levels, ruling out effects larger than 2% two years after. These null results hold for intensive users, early adopters, workplaces with substantial investments, workers reporting large gains, flexible-pay occupations, and early-career jobs. Adoption is linked to occupational switching and task restructuring, but without net changes in hours or earnings. Our findings challenge narratives of imminent disruption from Generative AI.

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1.2 Register Data

We use several administrative registers at Statistics Denmark. Our matched employer-employee data come from the *Employment Statistics of Employees* (BFL), which records earnings, hours, occupation, and industry for all job spells in Denmark on a monthly basis from 2008 onward. This register is compiled by the Danish tax authorities and subsequently harmonized by Statistics Denmark. We complement this with demographics data on individuals from the *Population Register* (BEF), and wealth information from the *Personal Wealth Register* (FORMPERS). Finally, we draw firm financial data (e.g., annual revenue and employment) from the *Firm Statistics* (FIRM) Register. Because our survey was sent to workers identified in the register data by their (deidentified) social security numbers (*pnr*), all respondents can be matched to the register data. While labor market data in the BFL register are updated continuously, we only have firm financial data up to 2022. For that reason, we focus our effect analyses on labor market outcomes.

1.3 Survey Sample

We invited 115,000 workers to participate in each of our survey rounds in 2023 and 2024. The registers at Statistics Denmark enabled us to target these invitations by occupation and workplace. In particular, for each of our 11 occupations, we conducted a workplace-based sampling procedure, first drawing a random set of workplaces within each occupation and then sampling all relevant workers at these workplaces. This sampling procedure maximizes the statistical power of the workplace-level analyses while keeping our sample representative. Appendix B.1 details the sampling protocol. We sent three reminders per survey round, two by e-mail and one by text. The invitation letters are in Appendix I. We received about 25,000 valid and complete responses to each survey. Appendix B.2 details our survey response rates. While our main analysis focuses on responses from the 2024 round, we use the 2023 round to examine early adopters.